Venkata Lakshmi Parimala Pasupuleti

AI Software Engineer · LLM Applications · RAG Pipelines · Prompt Engineering · Full-Stack Python · Docker · RabbitMQ · CI/CD · Generative AI · AWS · Agile

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PROFESSIONAL SUMMARY

AI Software Engineer with 8 years of full-stack production experience and two shipped Generative AI systems - AgentProbe and LeaseI - built on the same LLM application stack this role targets: RAG pipelines, prompt engineering, LLM orchestration, and retrieval-augmented generation. AgentProbe is a production RAG evaluation platform with a 5-metric quality engine, LLM-as-judge scoring, and CI-integrated regression gating across 1,240+ test cases. LeaseI is an 8-agent LangGraph system with multi-source RAG, HITL escalation, and 93.6% precision on compliance test cases. Backend depth spans Python, FastAPI, Docker, RabbitMQ (messaging), PostgreSQL, and AWS (S3, EC2, RDS) - all preferred qualifications in this role. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Generative AI: LLM Prompt Engineering, RAG Pipelines, LangGraph, LangChain, Agentic AI, LLM Orchestration, ChromaDB, Hugging Face (API), OpenAI API, Anthropic API, Model Evaluation, Hallucination Detection, HITL, MCP, PyTorch, scikit-learn, BERT, spaCy, NLP

Full-Stack & Backend: Python, JavaScript, TypeScript, React, HTML, JSON, REST APIs, FastAPI, Flask, Node.js, SQL, PostgreSQL, MySQL, Oracle, Docker, Microservices, RabbitMQ, Redis, AWS (S3, EC2, RDS, Glue), CI/CD (GitHub Actions), Linux, Git

Engineering & Data: Agile/Scrum, Security Best Practices (PHI compliance, HIPAA, Libsodium), NumPy, Pandas, PySpark, Feature Engineering, Statistical Modeling, Tableau, Automated Testing, Observability

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Designed and developed AI application using LangGraph and RAG - LLM prompt engineering for NLP ingestion, retrieval-augmented generation over distributed data sources, and validation guardrails enforcing compliance before delivery.
- Implemented generative AI pipeline with Hugging Face-compatible embeddings; prompt engineering for response grounding; CI/CD-integrated automated testing of AI outputs.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (Python, PyTorch, scikit-learn, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$; full pipeline from data ingestion through feature engineering, model optimization, and statistical validation across 190+ countries.
- Mentoring 600+ students in Python, SQL, and AI/ML concepts; designed dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Full-stack development across 20+ scalable production modules** - Python/FastAPI backends, JavaScript/TypeScript, REST APIs, PostgreSQL/MySQL/Oracle databases; 99.5% uptime at 100K+ monthly active users across 120+ Agile sprints.
- **Microservices, messaging, and cloud infrastructure** - Docker containerization (50% scalability improvement); RabbitMQ messaging broker (async processing, retry logic, dead-letter queues); AWS S3, EC2, RDS at production scale; CI/CD via GitHub Actions.
- **Security best practices and compliance** - 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents); secure coding standards across regulated SaaS platform; 200+ SQL query optimizations (30% performance gain).
- Collaborated with product, design, and architecture teams to translate ambiguous requirements into production-ready solutions; mentored 5+ engineers; 300+ code reviews.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL); scalable data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Generative AI Application & RAG Evaluation Platform GenAI · RAG

Python · FastAPI · LangGraph · ChromaDB · Hugging Face · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Production generative AI application: 15-endpoint FastAPI service, RAG pipeline with LLM prompt engineering, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression gate, and multi-provider LLM routing. Evaluates AI outputs across 1,240+ test cases - zero regressions shipped to production.

Leasel: 8-Agent RAG Application with LLM Orchestration RAG · Agentic

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Scalable generative AI application: 8-agent LangGraph system with prompt engineering, multi-source RAG, conditional routing, and HITL escalation when model confidence is below threshold. 93.6% precision, 87.9% F1 across 1,240 compliance test cases; CI quality gate with telemetry-driven scoring.

AI on FHIR: NLP-Driven Medical Records Query System NLP · Healthcare

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application (Python + React) using LLM inference and BERT embeddings to translate natural-language queries into FHIR API calls - reduced retrieval time ~60%; HIPAA-compliant RBAC across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017